SECURITY AWARENESS PROGRAM

Phishing Awareness Training

Recognizing threats, identifying social engineering tactics, and executing defensive protocols.

Prepared by: Muskan Kumari

B.Tech CSE in Cyber Security | Rungta College of Engineering and Technology

OVERVIEW

Introduction to Phishing

3.4B

Daily phishing emails sent worldwide

90%

Cyber breaches originating from email

#1

Initial vector for enterprise ransomware

CORE CONCEPT

What is Phishing?



Deceptive Fraud

Attacks masquerading as trusted institutions.



Credential Theft

Harvesting corporate logins and passwords.



Malware Delivery

Infecting endpoints with ransomware payloads.

ATTACK VARIANTS

Types of Phishing Attacks

Email Phishing

Mass untargeted email lures.

Spear Phishing

Tailored individual targets.

Whaling

High-level C-suite targets.

Smishing & Vishing

SMS and voice call scams.

EXECUTION LIFECYCLE

How Phishing Attacks Work

STEP 01

Reconnaissance

Gathering victim intelligence from OSINT sources.

STEP 02

Weaponization

Cloning portals and setting up spoofed domains.

STEP 03

Delivery

Dispatching malicious email lures to inbox.

STEP 04

Exploitation

Victim inputs login credentials or runs malware.

DETECTION SKILLS

Recognizing Phishing Emails

- **Sender Domain Mismatch**
  From address differs from real corporate URL.
- **Generic Greetings**
  Salutations like "Dear Valued Customer".
- **Suspicious Attachments**
  Unexpected .zip, .xlsm, or double-extension files.



WEB SECURITY

Identifying Fake Websites

Fake Malicious Portal

http://paypa1-security-check.com

Typosquatting substitution ("1" instead of "l").

Unencrypted HTTP connection without valid certificate.

Official Legitimate Portal

https://www.paypal.com/signin

Verified primary corporate root domain.

Encrypted HTTPS SSL padlock connection active.

PSYCHOLOGICAL TRIGGERS

Social Engineering Techniques

Artificial Urgency

Bypassing critical thinking via quick deadlines.

Authority Pressure

Impersonating executive leadership or IT.

Fear & Intimidation

Threats of account suspension or penalties.

Greed & Curiosity

Promising unexpected bonuses or rewards.

CASE STUDIES

Real-World Phishing Example



Tech Giants Fraud ($100M BEC Loss)

Spoofed vendor email accounts sent forged invoices over two years.

Critical Infrastructure Shutdown

Leaked employee credentials enabled entry for pipeline ransomware.

QUICK REFERENCE

Common Warning Signs

Unsolicited Links

Embedded buttons masking destination URLs.

Poor Grammar

Awkward phrasing and spelling mistakes.

Unusual Requests

Demands for urgent gift cards or wire transfers.

Coercive Tone

Threatening swift disciplinary or legal action.

DEFENSE TACTICS

Best Practices to Stay Safe

Enable MFA

Multi-Factor Authentication stops stolen password access.

Hover Before Clicking

Inspect full URL target destination in browser status bar.

Verify Out-of-Band

Confirm unexpected requests via known phone numbers.

Keep Systems Updated

Patch software to close vulnerability exploits.

INCIDENT RESPONSE

What to Do After Clicking a Phishing Link

01

Disconnect Network

Unplug Ethernet or disable Wi-Fi instantly to stop malware spread.

02

Reset Passwords

Change credentials on clean device and revoke active sessions.

03

Report to IT / SOC

Alert security team with exact link and email headers.

04

Run Antivirus Scan

Execute full endpoint security scan to isolate threats.

KNOWLEDGE CHECK

Interactive Quiz Scenario

Scenario: You receive an email from "IT HelpDesk" (support@company-login-verify.com) requesting password verification within 30 minutes. What should you do?

A Click the verification link immediately to prevent lock-out.

B Do not click; inspect headers, verify domain, and report to IT Security.

C Reply to the sender asking if the email is legitimate.

ANSWER ANALYSIS

Quiz Answer & Key Lessons

Correct Answer: Option B

Inspect domain, avoid links, and report directly to IT Security.

Why Option A is Wrong

Urgency is a manipulation tactic; clicking leads to credential theft.

Why Option C is Wrong

Replying validates your active email address to malicious attackers.

Conclusion: Stay Alert, Stay Secure

Security is a shared responsibility. When in doubt: Stop, Think, Verify, and Report.

Report Suspicious Emails: security@company.com

Image Sources



https://imagedelivery.net/KxWh-mxPGDbsqJB3c5_fmA/67cdd0d2-cea2-4c0d-ee46-664d4f5b0600/public

Source: keepnetlabs.com



https://media.easy-peasy.ai/27feb2bb-aeb4-4a83-9fb6-8f3f2a15885e/0385edb4-f72c-4b62-99d0-09504d2805bd_medium.webp

Source: easy-peasy.ai